Higher National Diploma in Quantity Surveying

Second Year, 1st Semester Examination – 2017

QS 21023 – Building Economics 1

Instructions for candidates:
Answer Four Questions out of Six
All questions carry equal marks.
Calculators are allowed.

No of pages : 04
No of questions: 05
Time: Three (03) hours

Q1

I. Explain briefly over view of the Construction industry in Sri Lanka. What are main Ministries directly involved to construction industry. (05 marks)

II. What are the factors affecting for delaying construction projects? (05 Marks)

III. Who are the Direct and Indirect Stakeholders in construction industry and explain the duties of direct stakeholders in Construction projects? (05 Marks)

IV. Name the five regulatory bodies of construction industry and briefly explain the duties of two of them. (05 Marks)

V. Professionals are the Key persons help to success a project. Name five Professionals actively participate in civil engineering projects and explain the duties of two of them (05 Marks)

Q2

CIDA (Previously named as ICTAD) is a main regulatory body of Construction industry to guide the Clients, Consultants and Contractors towards sustainability of constructions in the country. One of the major activities of CIDA is National Registration and Grading scheme for Contractors. It is helped both clients and contractors for mutual benefits. CIDA is actively participating for Grading of Contractors.

I. What are the specialized fields, contractors have to register under CIDA. (05 Marks)

II. What are the main criteria considered by CIDA for Registration of Contractors? (05 Marks)

III. There is a lack of Skill workforce for construction industry. Explain what are the reasons for that and what steps can be taken by Government to increase the skill work force. (05 Marks)

IV. Cost indices are very much important to predict the estimated cost of construction related works. What do you mean by "cost indices" and Explain how do you use the cost indices for predict the cost estimate for construction projects? (05 Marks)

V. Government policies relative to the construction industry are very much important to uplift the construction industry in the country. Propose five policies Government can implement to uplift the construction industry (05 Marks)

Q3

RIBA/SLIA Plan of work is the stages of proceeding from feasibility stage to completion stage of the project

I. Explain Why it is important to follow the RIBA/SLIA plan of work for formulation of Projects (05 marks)

II. What are the stages of RIBA/SLIA related to Pre Contract stage? Explain the stages of Project Appraisal and conceptual design. (05 Marks).

III. Name three feasibilities which have been considered in Projects in Appraisal stage. Briefly Explain two of them (05 Marks)

IV. What do you mean by EIA report for a project? Briefly explain the contents of an EIA report. (05 Marks)

V. What are the RIBA/SLIA plan of work stages of Quantity Surveyor involved under the Pre and Post Contract stages? (05 marks)

Q4 **National Income** or **Gross Domestic Product (GDP)** is an indicator of National Development of the country. The other indicators are Literacy Ratio, Infant Mortality Ratio, Per capita income, Life Expectancy and Employments ratio. Developed countries show high values of these indicators. Economic growth rate is also the indicator for measuring National Development in a Country. Micro Economic and Macro-economic both influence GDP.

I. What is meant by GDP? What are three approaches in calculating GDP in a Country? (05 Marks)

II. Briefly explain the meaning of following indicators in a country
   a. Literacy Ratio
   b. Infant Mortality Ratio
   c. Per capita income,
   d. Life Expectancy
   e. Employments (05 Marks)

III. What are the main sectors considered in GDP? (05 Marks)

IV. Explain the main features are considered in Macro Economics and Micro Economics. (05 Marks)

V. Briefly explain the economic growth in a country (05 Marks)

Q5

I. Explain the different kind of estimates. (05Marks)

II. You are required to submit a rough estimate for Warehouse construction of 6m high 3000 sqm in 2017. Cost data and cost indices relative to 2010 and 2017 are given in table below with relative to the civil construction. Estimate following

   a. Unit cost per cubic meter in warehouse constructed in the year 2010 (03 marks)
   b. Unit cost per cubic meter in warehouse proposed to be constructed in 2017 (03 Marks)
   c. Total cost of proposed warehouse to be constructed in 2017 (04 marks)

Year	Floor Area in sqm	Height in m	Cost in Rs	Unit Cost per cum	Construction Cost index
2010	2500	8	2,500,000.00	a	120
2017	3000	6	c	b	165

III Domestic two stories Building with floor area in ground floor is 150 sqm and first floor is 80 sqm. Unit rate per sqm is Rs 50,000/= for ground floor and 60,000/= for first floor. Reduction of floor area for Stairs and Courtyard is 10% for ground floor and 5% for first floor respect to total area of ground floor and first floor. Following cost are related as percentage of Preliminary cost Estimates.

Architectural and structural design cost is 2%.
Utility service cost (Water, Electricity and Sanitary) cost is 6%
Construction supervision cost is 3%

a) Calculate the effective Plinth area for Ground floor
b) Calculate the effective Plinth area for First floor
c) What is the Plinth area cost for Ground floor?
d) What is the plinth area cost for First Floor?
e) What is the Total Plinth area Cost for the Building?
f) Estimate the Architectural and structural Design cost
g) Estimate the Utility Service Cost
h) Calculate the total estimate cost of the Building
i) What will be the accuracy of preliminary estimate compare with detail Estimate (BOQ)?
j) Why Rough or Preliminary estimate is important in Client point of view?
(10 Marks, Each has equal mark)